MIL-T-28954(MC)
22 July 1970
SUPERSEDING
See Section 6.

MILITARY SPECIFICATION

TEST SET, RADIO: AN/GRM-21

1. SCOPE

1.1 This specification covers the requirements for a Radio Test Set, AN/GRM-21 designed to permit testing and servicing components of Radio Set, AN/TRC-75, under operating conditions.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-529 - Enamel, Alkyd, Semigloss.

MILITARY

MIL-T-704 - Treatment and Painting of Materiel.
MIL-P-15024 - Plate, Identification-Information and Marking for Identification of Electrical, Electronic, and Mechanical Equipment.
MIL-E-17555 - Electrical and Electronic Equipment, Accessories and Repair Parts, Packaging and Packing of.
MIL-T-21200 - Test Equipment for Use With Electronic and Fire Control Systems, Generals, General Specification for

STANDARDS

FEDERAL

FED-STD-595 - Colors.

FSC 6625

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MILITARY

| | | |
|---|---|---|
| MIL-STD-129 | - | Marking for Shipment and Storage. |
| MIL-STD-454 | - | Standard General Requirement for Electronic Equipment. |
| MIL-STD-810 | - | Environmental Test Methods. |
| MIL-STD-831 | - | Test Reports, Preparation of. |

DRAWINGS

MARINE CORPS

| | | |
|---|---|---|
| 522-1797-000 | - | Collins Radio Co. (Mfr. Code 13499) Drawings for Test Set, Radio, AN/GRM-21. |

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), as soon as practical after award of contract, the contractor shall furnish a first article to determine compliance with this specification. No additional test sets shall be delivered prior to approval of the first article.

3.2 Parts and materials. The parts and materials for the test set shall be the same quality, grade and type as those used in the Government loaned model (see 3.9). The interchangeability of assemblies shall meet the requirements of MIL-T-21200. The use of parts and materials, other than those required by MIL-T-21200, shall be evaluated and approved by the contracting officer.

3.3 Design and construction. The design and construction of the test set shall be the same as the Government loaned model (see 3.9). The test set shall consist of one each of the following: adapter cable assemblies, [illegible]s, dummy loads, hoses, mountings, and wiring harnesses:



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FAN, VANEAXIAL MD-277/U
MOUNTING MT-1864/GRM-10
MOUNTING MT-1868/GRM-10
MOUNTING MT-1865/GRM-10
MOUNTING MT-1866/GRM-10
MOUNTING MT-1867/GRM-10
MOUNTING MT-1869/GRM-10
WIRING HARNESS CX-4161/GRM-10
ADAPTER, TEST MX-2244/GRM-10
WIRING HARNESS CX-4162/GRM-10
WIRING HARNESS CX-4164/GRM-10
WIRING HARNESS CX-4165/GRM-10
WIRING HARNESS CX-4166/GRM-10
WIRING HARNESS CX-4167/GRM-10
DUMMY LOAD, ELEC. DA-173/GRM-10
INTERCONNECTING BOX J-1125/GRM-21
MOUNTING MT-2267/GRM-21
MOUNTING MT-2268/GRM-21
CABLE ASSEMBLY, R-F CG-1607/U
CABLE ASSEMBLY, R-F CG-1727/U
DIAGRAM, INTERCONNECTING WIRING AN/GRM-21
CABLE ASSEMBLY, R-F CG-1608/U
ADAPTER, CONNECTOR UG-29B/U
DUCT, AIR 1-1/2 dia. CUFF.
DUCT, AIR 1-5/8 dia. CUFF.
DUMMY LOAD, ELECT. DA-229/GRM-21
MOUNTING MT-2169/U
CLAMP-HOSE
LEAD ASSY. ELECTRICAL
EQUIPMENT SPECIFICATION
PRODUCTION TEST REQUIREMENT
PLATE-COVER, TUNING

3.4 Performance. The test set shall meet the following performance requirements:

3.4.1 Environmental conditions. The test set shall be capable of operating under the following environmental conditions:

(a) Temperature. -65°C to +85°C.

(b) Humidity. 0-90° relative.

(c) Shock. As encountered when dropped from a height of 4 inches onto a wooden bench.



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3.4.2 Power input. The power input required for operation of the test set shall be 120 volts (V) ±5 percent, 400 ±5 hertz, 3 phase and 28 ±1 V, 5 amperes.

3.5 Dimensions. The dimensions of the test set shall be within limits shown in Drawing No. 522-1797-000.

3.6 Weight. The weight of test set shall be the same as the Government furnished model (see 3.9).

3.7 Treatment and painting. Treatment and painting shall be in accordance with MIL-T-704. All metal parts shall be painted with enamel conforming to TT-E-529. Color shall match gray No. 26329 of FED-STD-595.

3.8 Identification marking. Identification plates shall conform to MIL-P-15024 and shall be the same as the plates on the Government loaned model, except for data relative to specific contract. The contractor shall furnish identification plates on each item of equipment.

3.8.1 Plug identification. All plugs shall be identified with the same designation and placed in the same location as the Government loaned model.

3.8.2 Symbol numbers. Reference symbol numbers shall be the same as on the Government loaned model and in accordance with Drawing No. 522-1797-000.

3.9 Government loaned property. The Government will loan one Radio Test Set, AN/GRM-21, and two Radio Sets, AN/TRC-75, for use as models for the duration of the contract. The Government loaned test set may be disassembled and tested in any way deemed necessary by the contractor to determine construction methods and techniques. Upon completion of the contract, the Government loaned models shall be returned to the Government in the same condition in which they were received; the test set may be replaced by the contractor with production equipment.

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454.

4. QUALITY ASSURANCE PROVISIONS



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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3)
(b) Acceptance (see 4.4)
(c) Sample (see 4.5)

4.3 First article inspection. The first article inspection shall be conducted on one first article test set which is representative of the production equipment. The first article inspection tests shall consist of all tests listed under First Article in Table I. The tests shall be performed in the sequence listed. All tests shall be witnessed and certified by the cognizant Government inspector.

4.3.1 First article test report. The first article test report shall be prepared in accordance with MIL-STD-831.

4.3.2 Approval of first article. Approval, conditional approval, or rejection of the first article shall be made in writing from the contracting officer after receipt of the first article test report.



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Table I - Tests

| | Test Para. | Reqt. Para. | Accep-tance | Sample | First Article |
|---|---|---|---|---|---|
| Visual inspection | 4.7.1 | 3.2<br>3.3<br>3.5<br>3.6<br>3.7 | X | X | X |
| Electrical tests | 4.7.2 | 3.4.2 | X | X | X |
| Low Temperature Tests | 4.7.3.1 | 3.4.1 | | | X |
| High Temper-ature Tests | 4.7.3.2 | 3.4.1 | | | X |
| Humidity Test | 4.7.3.3 | 3.4.1 | | | X |
| Shock Test | 4.7.3.4 | 3.4.1 | | X | X |
| High Potential Test | 4.7.3.5 | | | | X |
| Weight | 4.7.4 | 3.6 | | | X |



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4.4 Acceptance inspection. Acceptance inspection tests specified in Table I shall be performed on each test set submitted for acceptance under the contract. The contractor shall furnish acceptance records in duplicate with each test set shipped. In addition, the contractor shall maintain records of all acceptance inspection tests. These records shall be made available to the cognizant Government inspector.

4.5 Sample inspection. One out of each 10 test sets produced shall be subjected to tests specified in Table I. Test sets submitted for sample tests shall be selected at random by the cognizant Government inspector and shall have passed the acceptance inspection test.

4.6 Test procedures. Test procedures for all three types of inspection specified herein shall be submitted to the contracting officer for approval before testing. The procedures shall be in accordance with the requirements for each inspection type as specified herein. Testing shall not be performed for record until approval of the test procedures is obtained from the procuring activity.

4.7 Inspection and tests. Unless otherwise specified herein, each test shall be conducted under normal ambient conditions. Normal ambient conditions shall be a temperature of $+25^{\circ} \pm 10^{\circ}C$, an altitude of 0 to 5,000 feet, and a relative humidity of 0 to 90 percent.

4.7.1 Visual inspection. The test set shall be visually inspected to determine correctness of materials, construction, color, finishes, marking and identification, workmanship, completeness, that all components fit in their respective position, electrical and mechanical connections, and general appearance.

4.7.2 Electrical tests. The set shall be connected, utilizing components of Government loaned Radio Set, AN/TRC-75, and operated in accordance with instructions furnished with Radio Set, AN/TRC-75. Communications with additional Radio Set, AN/TRC-75, shall be established at frequencies throughout frequency range of equipment.



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4.7.3 Service condition tests. The test set shall be visually inspected, both internally and externally, immediately before and after each of the following service condition tests. Any signs of corrosion, damaged parts, or other defects shall be cause for rejection of the test set.

4.7.3.1 Low temperature. The test set shall be tested in accordance with MIL-STD-810, method 502, procedure I. Storage temperature shall be -65°C.

4.7.3.2 High temperature. The test set shall be tested in accordance with MIL-STD-810, method 501, procedure I. Storage temperature shall be +85°C.

4.7.3.3 Humidity. The test set shall be tested in accordance with MIL-STD-810, method 507, procedure II. Electrical tests of 4.7.2 shall be conducted at the end of each cycle.

4.7.3.4 Shock. The test set shall be tested in accordance with the requirements of MIL-T-21200. The test set shall meet the requirements of test in 4.7.2 upon completion of the shock test.

4.7.3.5 High potential. The test set shall be subjected to 400 V root mean square high potential test in accordance with MIL-T-21200. The voltage shall be applied between connector pins and between connector pins and connector ground. Any current leakage or arcing will be cause for rejection.

4.7.4 Weight. The complete test set shall be weighed on a calibrated scale. The weight shall not exceed the weight of the Government loaned model.

4.8 Rejected units. Any test set which does not meet the full requirements of this specification shall be rejected. Before resubmittal, full particulars concerning previous rejections and corrective action taken to prevent recurrence shall be made available to the Government inspector.

5. PREPARATION FOR DELIVERY



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5.1 Preservation, packaging and packing. Preservation, packaging and packing shall be the level of shipment specified in the contract or order in accordance with MIL-E-17555 (see 6.2).

5.2 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The test set covered by this specification is intended to be used as a bench test service to test the Radio Set, AN/TRC-75.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number and date of this specification
(b) When first article is not required (see 3.1)
(c) Levels of preservation, packaging, and packing required (see 5.1).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions C-03, T-01, T-02, T-03, T-04 and T-10 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 First article approval. The first article shall be examined and tested for approval at the contractor's plant or at an independent commercial testing laboratory acceptable to the procuring activity. First article tests shall be witnessed by a Government representative of the procuring activity. Approval of the first article shall be by the contracting officer.

6.4 Supersession data. This specification includes the requirements of Marine Corps Purchase Description 61-15A, Test Set, Radio AN/GRM-21 dated 27 November 1961.

Preparing activity:
Navy - MC
Project No. 6625-N253



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